

FINAL PROJECT REPORT

1.Introduction

Online learning has become an important part of modern education, providing students with flexible access to learning resources, courses, and academic activities. However, understanding how students engage with online learning platforms and how their engagement relates to academic performance is important for improving learning outcomes.

The “**Online Learning Engagement and Student Performance Analysis**” project focuses on analyzing student data to identify patterns and relationships between **online learning engagement and academic performance**. The project uses **SQL** for data storage, cleaning, management, and analysis, and **Power BI** for creating interactive dashboards and visualizations.

The project involves collecting and preparing the dataset, performing **data quality checks and SQL-based analysis**, and presenting the results through meaningful **KPIs, charts, and graphs**. The dashboard helps users understand student engagement levels, performance patterns, and important trends within the available data.

1.1Project Overview

The project focuses on analyzing **online learning engagement and student performance** using data analytics. The dataset contains information related to students, their learning activities, engagement, and academic performance. SQL is used for **data storage, cleaning, querying, and analysis**, while Power BI is used to create **interactive dashboards and visualizations**.

The project aims to identify patterns such as:

- Student engagement and learning activity
- Academic performance and scores
- Attendance/participation patterns
- Relationship between engagement and performance
- Student-wise and overall performance analysis
- Key trends and insights that can help understand student learning behavior

The final outcome is an **interactive Power BI dashboard** supported by SQL-based data analysis, allowing users to easily understand student engagement and performance through charts, KPIs, and visual reports.

1.2. Objectives

- To analyze online learning engagement of students using available learning activity data.
- To evaluate student academic performance based on scores, grades, and other performance-related factors.
- To identify the relationship between student engagement and academic performance.
- To analyze student participation and learning patterns such as attendance, study activity, and course interaction.

- To identify trends and patterns in student learning behavior using SQL-based data analysis.
- To create interactive dashboards in Power BI for clear visualization of student engagement and performance.
- To generate meaningful insights that can help understand factors associated with better student performance.
- To support data-driven decision-making by presenting student performance and engagement information in an easy-to-understand format.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

With the increasing use of online learning platforms, students generate a large amount of data related to their learning activities, engagement, participation, and academic performance. However, it can be difficult to understand this data and identify patterns using traditional methods.

The **Online Learning Engagement and Student Performance Analysis** project aims to analyze student engagement and academic performance using **SQL and Power BI**. The project focuses on identifying patterns and relationships between students' online learning activities and their performance.

The analyzed data is transformed into meaningful **KPIs, charts, and interactive dashboards** to provide a clear view of student engagement and performance. This helps in understanding learning patterns and generating useful data-driven insights.

2.2. Project Proposal (Proposed Solution).

The **Online Learning Engagement and Student Performance Analysis** project proposes to analyze student learning activities and academic performance using **SQL and Power BI**. The project will use student-related data to examine engagement, participation, learning behavior, and performance.

SQL will be used for **data storage, cleaning, transformation, and querying**, while Power BI will be used to create **interactive dashboards and visualizations**. The analysis will identify important patterns and relationships between students' online learning engagement and their academic performance.

The proposed solution will provide meaningful **KPIs, charts, and reports** that make the data easier to understand and help users gain insights into student learning behavior and performance.

2.3. Initial Project Planning

- The initial planning of the Online Learning Engagement and Student Performance Analysis project involves defining the project requirements, data, tools, and analysis approach.
- Define the Project Scope – Analyze online learning engagement and student academic performance.

- Identify Data Requirements – Collect student information, learning activities, engagement details, and performance-related data.
- Select Tools – Use MySQL/SQL for database management and data analysis, and Power BI for visualization and dashboard creation.
- Data Preparation – Check the dataset for missing values, duplicate records, incorrect data types, and other data quality issues.
- Database Development – Create the required database tables and load the prepared data into MySQL.
- Data Analysis – Write SQL queries to calculate important measures and identify patterns in student engagement and performance.
- Dashboard Planning – Identify suitable KPIs, charts, filters, and visuals required for the Power BI dashboard.
- Final Analysis and Reporting – Present the findings through an interactive dashboard and summarize the key insights from the analysis.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan

The project will use student-related online learning and academic performance data to analyze engagement and performance patterns. The data will be collected, reviewed, and prepared before loading it into the SQL database.

- Collect relevant **student learning and performance data**.
- Include details related to **student information, engagement, learning activities, and academic performance**.
- Check the collected data for **missing values, duplicate records, and incorrect data types**.
- Clean and transform the data using SQL before analysis.
- Store the prepared data in **MySQL** for further querying and analysis.
- Connect the database data to **Power BI** for dashboard creation and visualization.

Raw Data Sources identifies

The raw data for this project consists of **student online learning and performance records** provided for the project analysis. The data may include fields such as:

- Student ID
- Student details

- Course/subject information
- Attendance or participation
- Online learning activity
- Engagement-related information

3.2. Data Quality Report

The collected student data was checked before performing analysis to ensure that it was accurate, complete, and suitable for SQL and Power BI analysis.

Data Quality Check	Observation	Action Taken
Missing Values	Some fields may contain null or blank values	Identified and handled appropriately
Duplicate Records	Duplicate student records were checked	Duplicate records were removed/verified
Data Types	Columns were checked for correct data types	Correct data types were assigned
Student ID	Student IDs were checked for consistency	IDs were verified and standardized
Invalid Values	Unusual or incorrect values were checked	Invalid values were corrected or excluded
Consistency	Similar values were checked for consistent formatting	Data was standardized
Completeness	Required fields were reviewed	Important fields were retained for analysis
Accuracy	Data values were reviewed for obvious errors	Errors were corrected where identified

Conclusion:

After performing the data quality checks, the dataset was prepared for **MySQL database analysis and Power BI dashboard development**. This helps ensure that the analysis and visualizations are based on reliable and consistent data.

3.3. Data Exploration and Preprocessing

The collected student data was explored and preprocessed to understand its structure and prepare it for analysis. The data was checked for missing values, duplicate records, incorrect data types, and inconsistent values.

Data Exploration

- Examined the **number of records and columns** in the dataset.
- Identified important fields such as **Student ID, course information, engagement, attendance, and performance scores**.
- Checked the **data types and unique values** of different columns.
- Analyzed basic patterns in student engagement and academic performance.

Data Preprocessing

- Checked and handled **missing or null values**.
- Identified and removed **duplicate records** where necessary.
- Corrected **incorrect or inconsistent data formats**.
- Converted columns into appropriate **data types** for SQL analysis.
- Checked for invalid or unusual values in numerical fields.
- Standardized the data to make it suitable for **SQL queries and Power BI visualization**.

After preprocessing, the cleaned dataset was stored in the **MySQL database** and used for further analysis and dashboard development in **Power BI**

4. Data Visualization.

4.1. Framing Business Questions

For the **Online Learning Engagement and Student Performance Analysis** project, the following business questions were framed to understand student learning behavior and academic performance:

- How many students are enrolled in the online learning program?
- What is the overall academic performance of the students?
- How does student engagement affect academic performance?
- What is the relationship between attendance and student performance?
- Which students show high or low levels of online learning engagement?

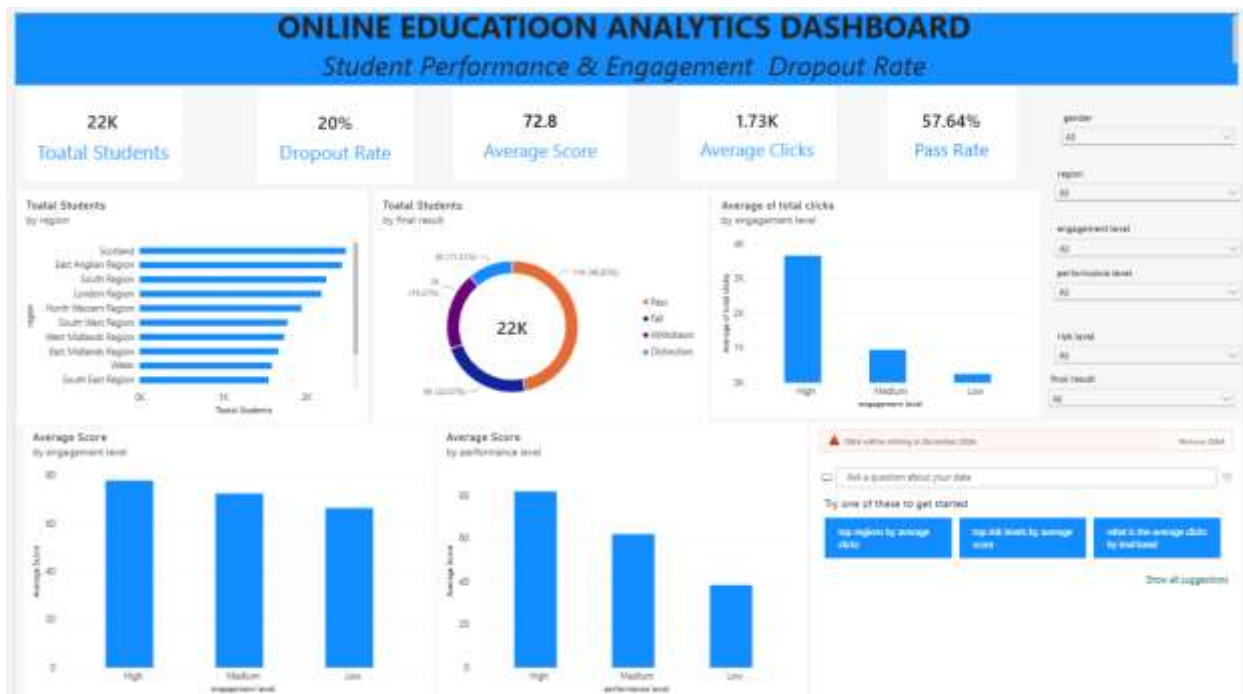
- Which courses or subjects have better or lower student performance?
- What learning patterns are associated with higher academic scores?
- Are there noticeable differences in performance based on students' level of participation?
- Which factors appear to have the greatest relationship with student performance?
- What insights can be obtained from student engagement and performance data to support better learning outcomes?

4.2. Developing Visualizations

The Online Learning Engagement and Student Performance Analysis project uses Power BI to convert the analyzed data into meaningful and interactive visualizations. The visualizations are designed based on the business questions identified in the previous stage.

5. Dashboard

5.1. Dashboard Design Screenshot with Insights



he visuals are formatted with appropriate titles, labels, and alignment to improve readability and provide an effective user experience.

Overall student performance: The dashboard covers approximately 22K students, with an average score of 73.07 and an average of 1.65K clicks.

Dropout remains significant: The 20.40% dropout rate shows that around one in five students are at risk of leaving their courses.

Pass rate needs improvement: The overall 57.27% pass rate indicates that a substantial proportion of students do not achieve a passing outcome.

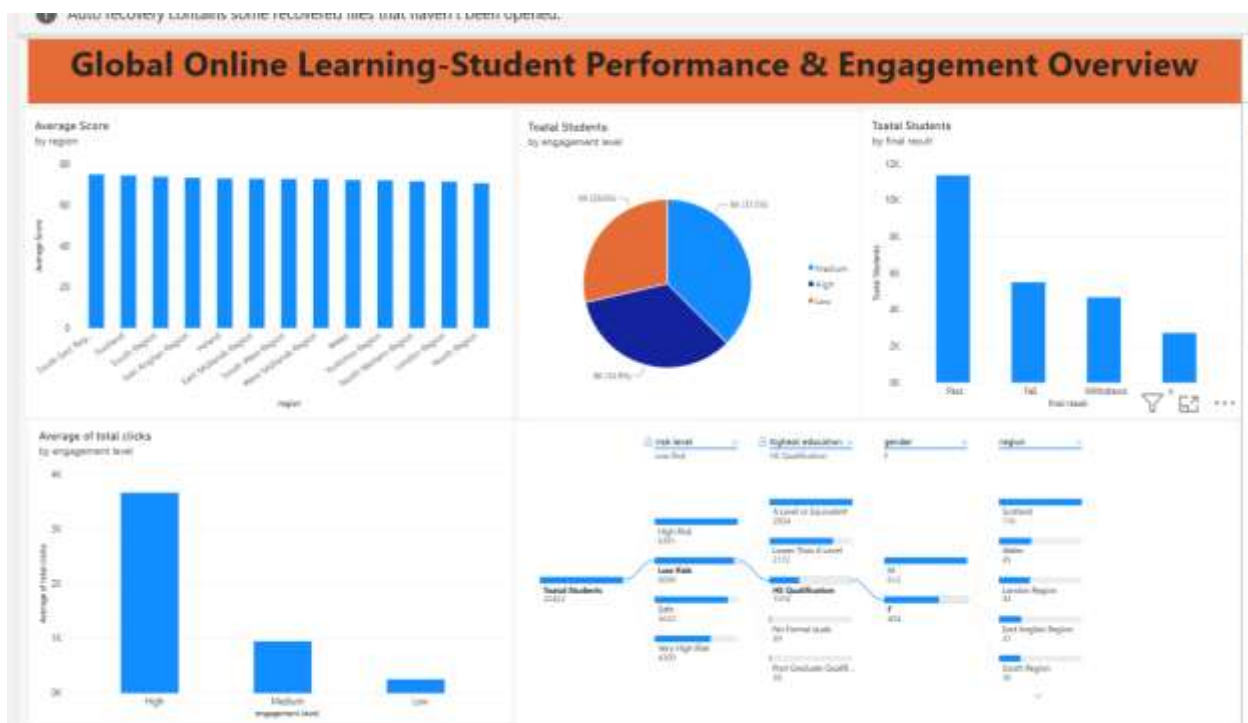
Regional differences are visible: Student distribution varies considerably across regions, with some regions having much larger student populations than others.

Engagement affects activity: Average clicks differ across engagement levels, showing that students demonstrate different levels of interaction with online learning content.

Performance varies by engagement: The dashboard shows differences in average scores across engagement levels, indicating that engagement is an important area for understanding student performance.

6. Report

6.1. Report Design Screenshot with Insights



Average Final Score by Region — Scotland Leads Regional Performance. Scotland records the highest displayed final-score value at 2,472, followed by East Anglian Region (2,424) and South Region (2,237), while North Region records the lowest value at 712, highlighting noticeable regional differences in student outcomes.

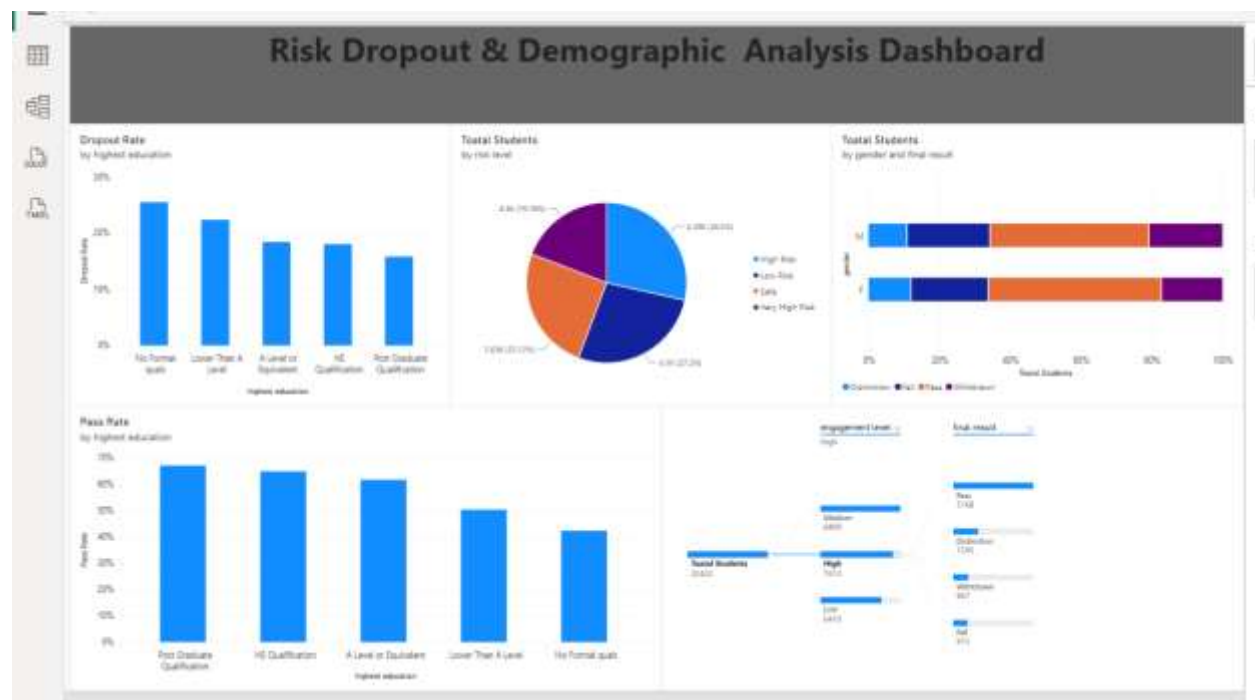
Students by Engagement Level — Engagement is Relatively Balanced. The student population is distributed across High, Medium, and Low engagement levels, with High engagement forming the largest group at approximately 37.5%, followed by Medium at 34.0% and Low at 28.5%.

Final Result Distribution — Pass is the Dominant Outcome. Pass has the highest number of students at approximately 10.4K, followed by Fail (5.0K), Withdrawn (4.5K), and Distinction (2.5K), indicating that successful completion is the most common outcome.

Average Total Clicks by Engagement Level — Higher Engagement Drives More Activity. High-engagement students generate the highest total clicks at approximately 27.6M, compared with 7.9M for Medium and 1.6M for Low engagement, showing a strong difference in online learning activity.

Decomposition Tree — High-Risk Students Form the Largest Risk Group. The decomposition analysis shows 6,391 High-Risk students, compared with 6,099 Low-Risk and 5,632 Safe students, helping identify segments that may require closer attention based on education, region, and gender.

Report 2



Dropout Rate by Highest Education — Lower Education Shows Higher Dropout. Students with No Formal Qualifications have the highest dropout rate at approximately 26%, while Post Graduate Qualification records the lowest at around 16%, indicating a clear difference across education levels.

Students by Risk Level — High-Risk Students Form the Largest Group. High Risk accounts for approximately 28.5% of students, followed by Low Risk (27.2%), Safe (25.1%), and Very High Risk (19.2%), highlighting the need for targeted student-support strategies.

Final Result by Gender — Pass is the Most Common Outcome. Both male and female students have Pass as the largest final-result category, with females showing a slightly higher pass count than males.

Pass Rate by Highest Education — Higher Education Has Better Outcomes. Post Graduate students have the highest pass rate at approximately 68%, while students with No Formal Qualifications have the lowest at around 43%, showing a positive relationship between education level and success.

Decomposition Tree — Medium Engagement Leads Student Outcomes. Medium-engagement students form the largest group with 8,400 students, followed by High (7,612) and Low (6,410). Within the Medium group, 4,158 students passed, making it the largest final-result segment.

7. Performance Testing

Performance testing is conducted to evaluate how efficiently the Online Learning Engagement and Student Performance Analysis system handles student-related data and generates analytical results. The testing focuses on the speed, responsiveness, accuracy, and smooth functioning of the dashboard and data processing operations.

The following aspects are considered:

- **Dashboard Loading Time:** Checks how quickly the Power BI dashboard and its visualizations load.
- **Data Processing:** Verifies that student engagement and performance data are processed efficiently.
- **Query Performance:** Checks whether SQL queries retrieve the required student data within a reasonable time.
- **Visualization Response:** Ensures that charts, graphs, filters, and slicers respond smoothly to user interactions.
- **Data Accuracy:** Verifies that the displayed totals, averages, performance measures, and engagement values match the underlying dataset.
- **Multiple Filters:** Tests whether the dashboard continues to respond properly when different filters and slicers are applied.
- **Overall Responsiveness:** Ensures that users can interact with the dashboard without unnecessary delays.

Expected Result: The dashboard should load properly, process the available student data efficiently, and provide accurate results with smooth interaction between filters, tables, and visualizations.

7.1 Utilization of Data Filters

Data filters are used in the **Online Learning Engagement and Student Performance Analysis** project to allow users to view specific portions of the student data. Filters make the Power BI dashboard interactive and help users analyze performance and engagement based on selected conditions.

Uses of Data Filters:

- **Student-wise filtering:** View the performance and engagement details of individual students.
- **Course-wise filtering:** Analyze student performance for a particular course.
- **Gender filtering:** Compare learning engagement and performance based on gender.
- **Age filtering:** Analyze students belonging to specific age groups.
- **Performance filtering:** Identify students with high, medium, or low performance.
- **Engagement filtering:** Analyze students based on their level of online learning engagement.
- **Time-based filtering:** View data for a specific date, month, or period.
- **Interactive analysis:** Filters allow users to change the displayed charts and tables dynamically.

Outcome: Data filters make the dashboard easier to use and help users quickly identify patterns, compare student groups, and understand the relationship between **online learning engagement and student performance**.

7.2 Number of Calculation Columns / Measures

- id student
- gender
- region
- highest education
- studied credits
- total clicks
- avg score
- engagement level
- performance level
- risk level
- pass flag

- dropout flag
- final result

7.3 Number of Visualizations

1. online educations analysis dashboard
2. Global Online Learning – Student Performance & Engagement Overview
3. Risk Dropout and Demographic Analysis

Report Visualizations: 5

1. **Average Score by Region** — *Average Score by Region*
2. **Student Distribution by Engagement Level** — *Total Students by Engagement Level*
3. **Students by Final Result** — *Total Students by Final Result*
4. **Average Total Clicks by Engagement Level** — *Average of Total Clicks by Engagement Level*
5. **Student Risk and Demographic Analysis** — *Risk Level, Highest Education, Gender and Region*

8.Conclusion

The **Global Online Learning – Student Performance & Engagement Overview** dashboard provides a clear analysis of student performance, engagement levels, learning activity, risk levels, and demographic characteristics. The visualizations help identify patterns in average scores, total clicks, engagement levels, and final results. Overall, the dashboard supports better understanding of student learning behavior and performance and can help educational institutions identify areas that require attention and improve online learning strategies.

9.Future Scope

The future scope of the **Online Learning Engagement and Student Performance Analysis** project includes:

- **Real-time data analysis** to monitor student engagement and performance continuously.
- **Predictive analysis** to identify students who may be at risk of poor performance.
- **Personalized learning recommendations** based on individual student activity and performance.
- **Advanced dashboards** with interactive filters and drill-down features.

- **Integration with learning platforms** to automatically collect updated student data.
- **Machine learning models** to predict final results and improve academic support.

10. Appendix

The appendix contains the supporting information, data, and materials used for the **Online Learning Engagement and Student Performance Analysis** project.

Appendix A – Dataset

The online learning dataset used for analyzing student engagement, performance, demographics, and final results.

Appendix B – Data Fields

Details of important fields such as student information, engagement level, total clicks, average score, risk level, education, gender, region, and final result.

Appendix C – Data Preprocessing

Details of data cleaning, handling missing values, checking data types, and preparing the dataset for analysis.

Appendix D – Power BI Dashboard

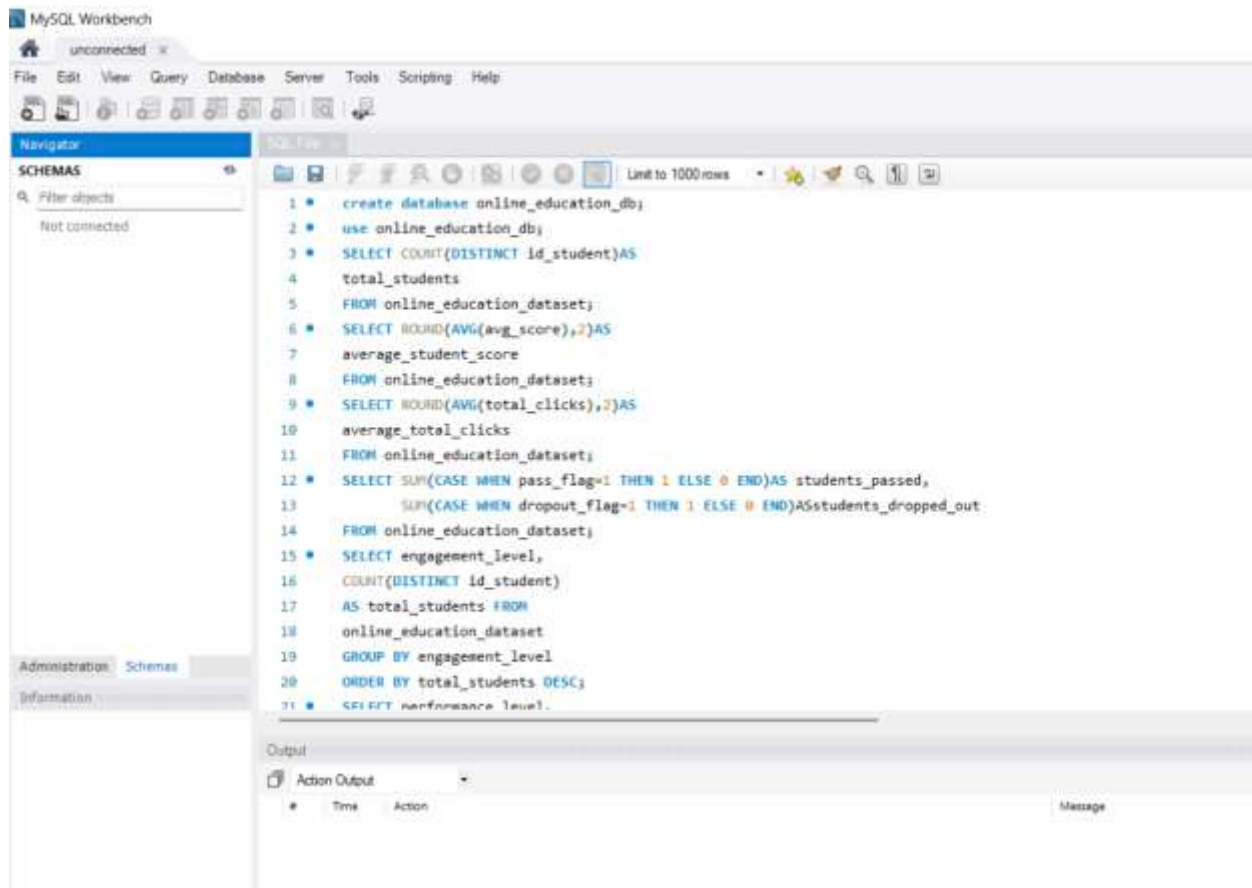
Screenshots of the completed dashboard and its visualizations.

Appendix E – Calculations and Measures

List of calculated columns and measures created for the analysis.

Appendix F – Report Visualizations

10.1 source code



10.2 Github Link Project Demo Link

<https://github.com/susmithadevalla241-wq/Online-Learning-Engagement-and-Student-Performance-Analysis>

